# Ensemble learning

INF-GRAM — Machine Learning for Healthcare

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# Session roadmap

# Why combine models?

A single model may be:

- unstable;
- overly sensitive to the training sample;
- too weak to capture the signal alone.

Ensemble methods try to improve predictive performance by combining multiple learners.

# Learning objectives

By the end of the lecture, you should be able to:

- distinguish voting, bagging, random forests and boosting;
- explain the intuition of each family;
- identify which methods mainly reduce variance;
- identify which methods build stronger learners sequentially.

# Voting: combine different models

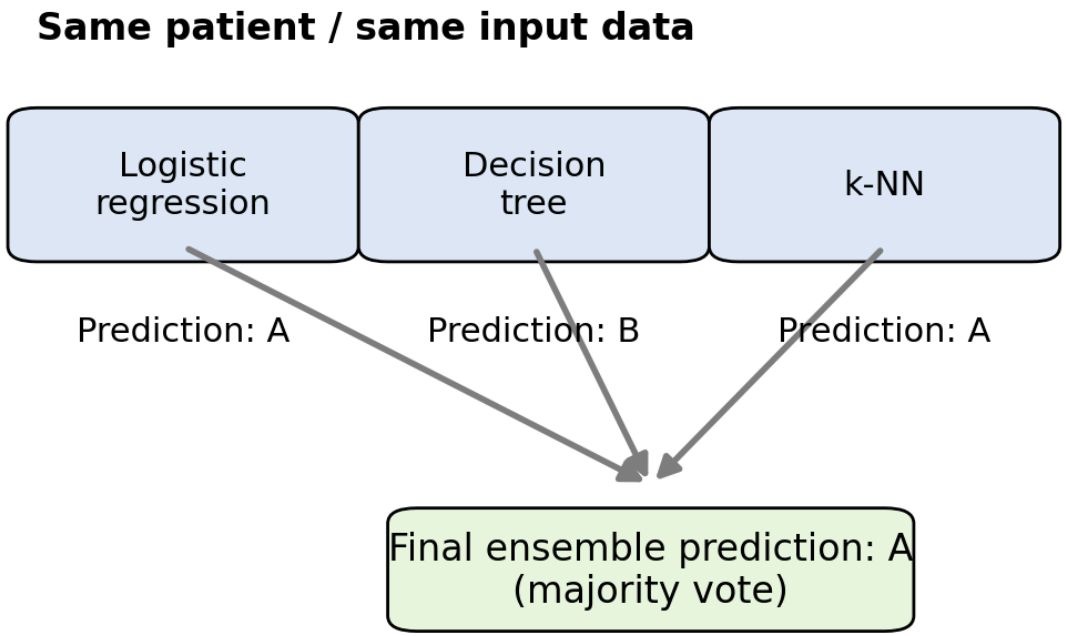
# Voting classifiers

The simplest ensemble combines different models trained on the same data.

Two common strategies are:

- **hard voting**: combine class labels by majority vote;
- **soft voting**: average predicted probabilities.

# Voting intuition



# When is voting useful?

Voting is useful when the component models make **different kinds of mistakes**. If they are too similar, the gain may be small.

# Bagging: reduce variance by resampling

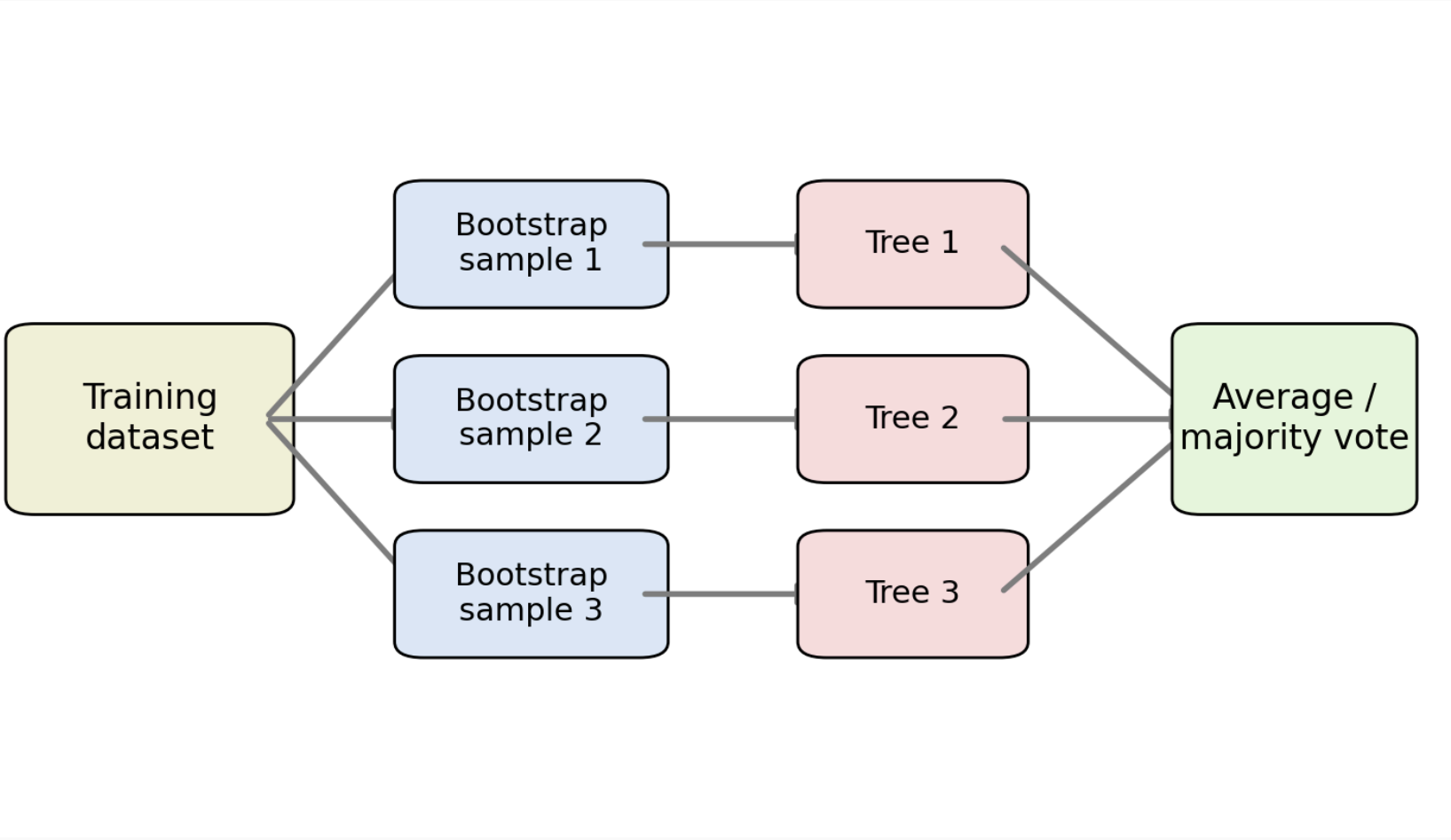
# Bootstrap aggregating

Bagging works in three steps:

1. draw many bootstrap samples from the training set;
2. fit one learner per sample;
3. aggregate their predictions.

This is especially effective for unstable learners such as decision trees.

# Bagging intuition



# What bagging mainly buys us

Bagging mainly reduces **variance**:

- each tree is noisy on its own;
- averaging stabilizes the prediction;
- the ensemble usually generalizes better.

# Random forests

# Random forest = bagging + feature randomness

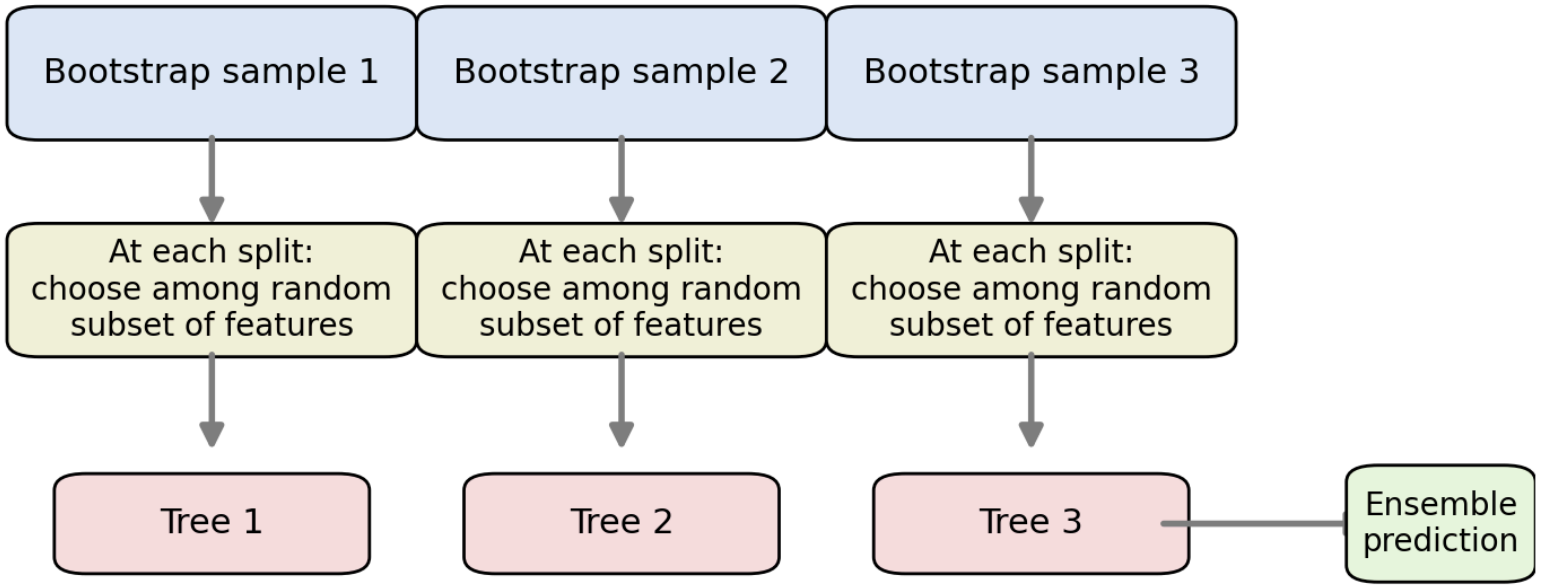
A random forest is a bagged ensemble of decision trees with one extra idea:

> at each split, each tree considers only a random subset of the predictors.

This helps decorrelate the trees.

# Random forest intuition

**Random forest = bagging of decision trees + random subset of features**



# Important tuning parameters

The most common hyperparameters are:

- `n_estimators` (number of trees);
- `max_features` (how many predictors are candidates at each split);
- tree-complexity parameters (`max_depth`, `min_samples_leaf`, ...).

# Why random forests are popular

Random forests are often strong default models because they are:

- robust;
- accurate;
- relatively easy to tune;
- less prone to overfitting than a single tree.

# Boosting

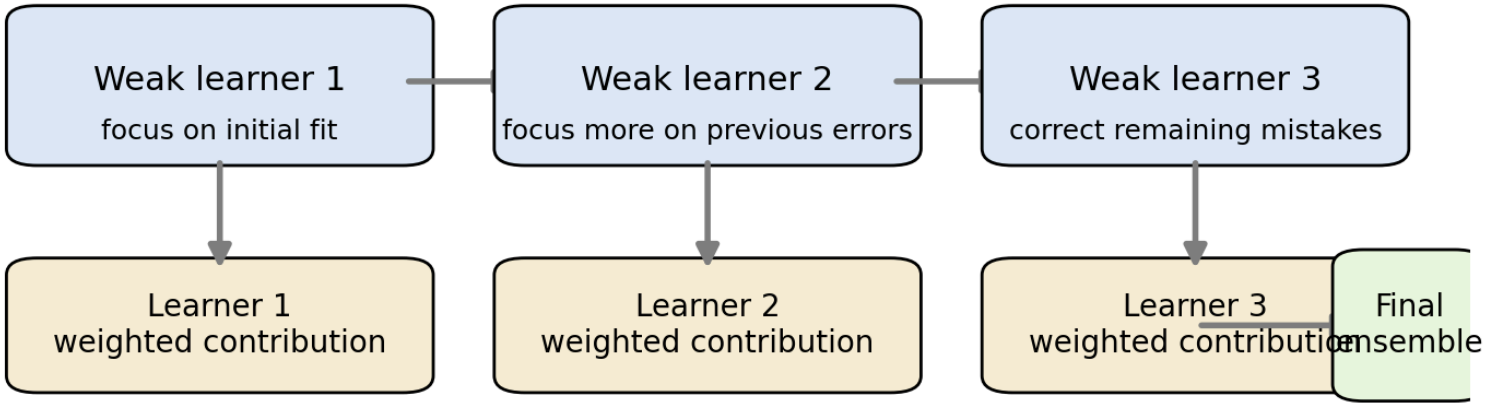
# The idea of boosting

Boosting does **not** fit all trees independently. It builds them **sequentially**.

Each new learner tries to improve the current ensemble by focusing on what is still hard to predict.

# Boosting intuition

**Boosting builds learners sequentially, each one improving the current ensemble.**



# Bagging vs boosting

## Bagging

- models are trained in parallel;
- mostly reduces variance;
- very natural for decision trees.

## Boosting

- models are built sequentially;
- can produce very strong learners;
- may overfit if tuning is poor.

# Putting the families together

# A compact summary

- **Voting**: combine diverse models.
- **Bagging**: stabilize unstable learners through resampling.
- **Random forest**: bagged trees + random feature subset.
- **Boosting**: sequential improvement of an ensemble.

# Practical message

Ensembles are powerful, but they still require:

- a proper train/validation/test workflow;
- sensible hyperparameter tuning;
- careful interpretation, especially in healthcare settings.

# Take-home messages

# What to remember from this lecture

- Ensembles often outperform single models because they are more stable or more expressive.
- Random forests are strong and robust general-purpose models.
- Boosting is often very powerful, but more sensitive to tuning.
- Better predictive performance never removes the need for careful evaluation.